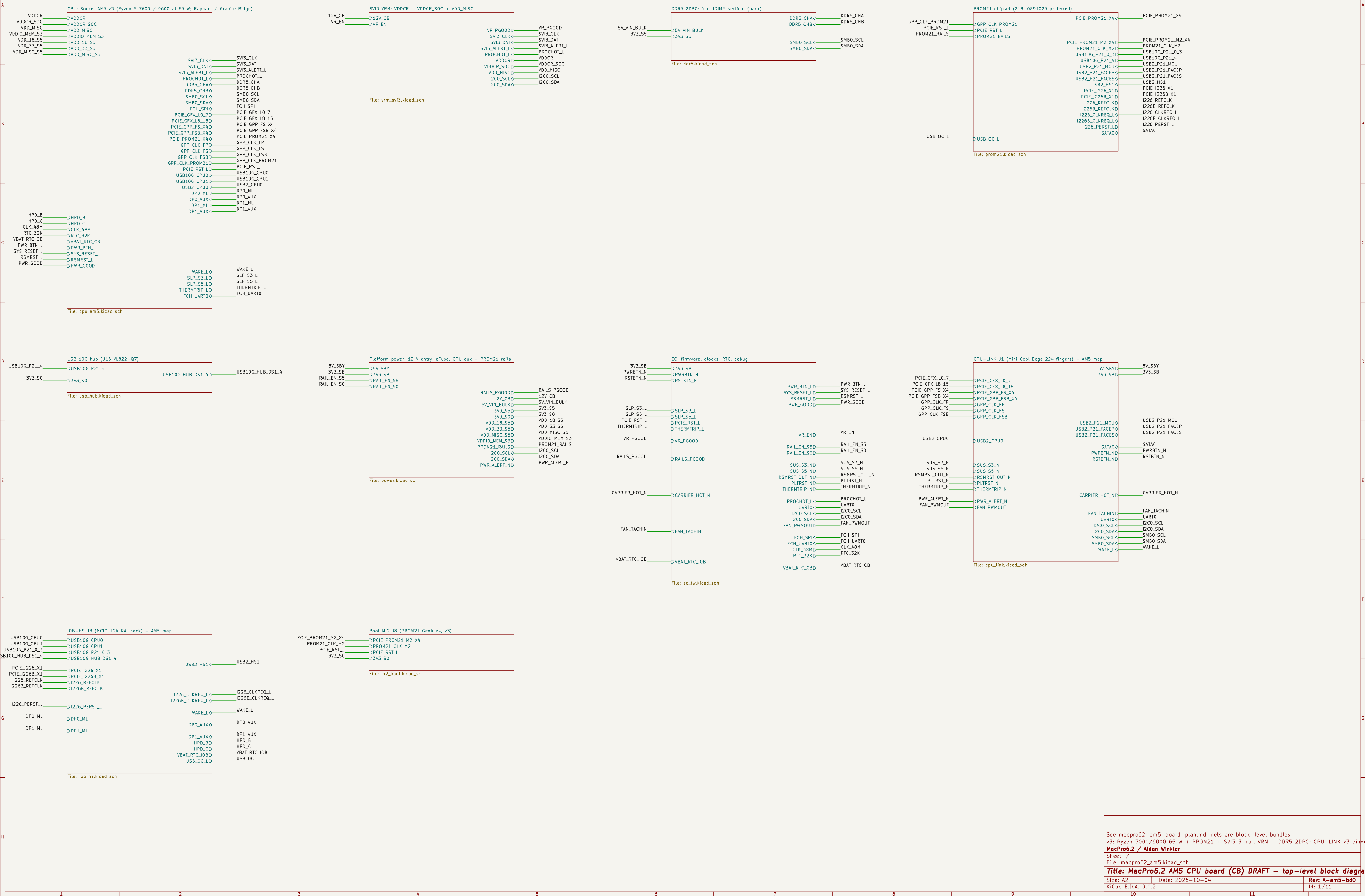


MP62 AM5 CPU BOARD rev A-am5 DRAFT – BLOCK DIAGRAM. Sheets are stubs (labels only). Same label name = same net (bundle).

Off-board: J1 CPU-LINK (card edge -> BP) | J3 IOB-HS MCIO 124 | J8 M.2 boot | LUG1/LUG2 12 V | J5 debug | J4 TPM. AMD pin names (NDA) replaced by functional names.



[illegible]





BLOCK STUB (rev A block diagram). Contents planned:

– J6/J7 (CH–A near/far) and J9/J10 (CH–B near/far): UMAX 90414 short–latch DDR5 UDIMM, same positions as the LGA1700 CB

– AM5 7000/9000: 2DPC 1R = DDR5–3600 (2R lower), 1DPC = DDR5–5200 (7000) / 5600 (9000) (AMD spec); daisy chain, populate far first

– Module PMIC from 5V_VIN_BULK (U7): SPD hub on SMBus0 (FCH), 3V3_S5 for SPD/HSA

– DDR5 40 ohm SE / 80 ohm diff stripline on L3/L8

Block stub: hierarchical labels + one non–BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /DDR5 2DPC: 4 x UDIMM vertical (back)/
File: ddr5.kicad_sch

Title: MP62 AM5 CPU board: DDR5 2DPC: 4 x UDIMM vertical (back)

Size: A4

Date: 2026–10–04

Rev: A–am5–bd0

KiCad E.D.A. 9.0.2

Id: 4/11

	1	2	3	4	5	6
A	PCIE_PROM21_X4◊<u>1</u> GPP_CLK_PROM21◊<u>2</u> PCIE_RST_L◊<u>3</u> PROM21_RAILS◊<u>4</u> PCIE_PROM21_M2_X4◊<u>5</u> PROM21_CLK_M2◊<u>6</u> USB10G_P21_0_3◊<u>7</u> USB10G_P21_4◊<u>8</u> USB2_P21_MCU◊<u>9</u> USB2_P21_FACEP◊<u>10</u> USB2_P21_FACES◊<u>11</u> USB2_HS1◊<u>12</u> PCIE_I226_X1◊<u>13</u> PCIE_I226B_X1◊<u>14</u> I226_REFCLK◊<u>15</u> I226B_REFCLK◊<u>16</u> I226_CLKREQ_L◊<u>17</u> I226B_CLKREQ_L◊<u>18</u> I226_PERST_L◊<u>19</u> SATA0◊<u>20</u> USB_OC_LD◊<u>21</u> PCIE_PROM21_X4 GPP_CLK_PROM21 PCIE_RST_L PROM21_RAILS PCIE_PROM21_M2_X4 PROM21_CLK_M2 USB10G_P21_0_3 USB10G_P21_4 USB2_P21_MCU USB2_P21_FACEP USB2_P21_FACES USB2_HS1 PCIE_I226_X1 PCIE_I226B_X1 I226_REFCLK I226B_REFCLK I226_CLKREQ_L I226B_CLKREQ_L I226_PERST_L SATA0 USB_OC_L BLOCK STUB (rev A block diagram). Contents planned: – U2 AMD Promontory 21 (B650/B850–class, single chip), FCBGA 19 x 19, ~7 W; 218–0891025 preferred over 218–0891018 (cost study sec. 10) – Uplink PCIe 4.0 x4 from the CPU; downstream: 10G USB x6 + 20G x1 class, USB2, SATA x4 class, PCIe 4.0/3.0 x1..x4 (B850 table) – Uses: 10G #0–3 -> J3 C3–C6, 10G #4 -> U16 VL822 hub (A1–A4), 10G #5 + 20G spare; USB2 -> J1 MCU/FACEP/FACES + J3 HS1 – PCIe x1 -> IOB i226 #1, x1 -> IOB ASM1182e (i226 #2 + AirPort); v3: free Gen4 x4 -> J8 M.2 boot; SATA0 -> J1 (BP OpenCore M.2) – Own SPI flash U17 (image from a retail board BIOS [Unverified]); REFCLK/PERST/CLKREQ for its downstream ports [Unverified] – Ballout, rails and strapping are AMD NDA -> open item					
B						
C						
D						Block stub: hierarchical labels + one non–BOM stub symbol (#BLKn, passive pins) so ERC sees the n Sheet: /PROM21 chipset (218–0891025 preferred)/ File: prom21.kicad_sch Title: MP62 AM5 CPU board: PROM21 chipset (218–0891025 preferred) Size: A4 Date: 2026–10–04 Rev: A–am5–bd0 KiCad E.D.A. 9.0.2 Id: 5/11
	1	2	3	4	5	6



BLOCK STUB (rev A block diagram). Contents planned:

- U9 RP2350 EC (as LGA1700): AM5 sequencing RSMRST_L -> PWR_BTN_L -> SLP_S5/S3 -> rails -> PWR_GOOD; VR_EN / VR_PGOOD
- Buffers to CPU-LINK (names unchanged): SUS_S3# = SLP_S3_L, SUS_S4-S5# = SLP_S5_L, PLTRST# = PCIE_RST_L, THERMTRIP#, RSMRST_OUT#
- LPT actuator on AM5: PPT/TDC/EDC via the SMU (Dasharo/openSIL interface TBD) – fallback fixed 65 W (or 45 W Eco) PPT limit in the BIOS + PROCHOT_L (CARRIER_HOT#)
- U10 FCH SPI ROM 32 MB (Dasharo coreboot + openSIL + PSP blobs), external programming; J4 SPI TPM header (fTPM default)
- Y1 48 MHz [Unverified] + 32.768 kHz; VBAT_RTC: IOB coin cell (J3 B26) diode-OR 3V3_S5 -> CPU RTC well; BT1 DNP; J5 debug (FCH UART, SWD, POST)

3V3_SB	3V3_S8
PWRBTN_ND	PWRBTN_N
RSTBTN_ND	RSTBTN_N
PWR_BTN_L	PWR_BTN_L
SYS_RESET_L	SYS_RESET_L
RSMRST_L	RSMRST_L
PWR_GOOD	PWR_GOOD
SLP_S3_LD	SLP_S3_L
SLP_S5_LD	SLP_S5_L
PCIE_RST_LD	PCIE_RST_L
THERMTRIP_LD	THERMTRIP_L
VR_EN	VR_EN
VR_PGOOD	VR_PGOOD
RAIL_EN_S5	RAIL_EN_S5
RAIL_EN_S0	RAIL_EN_S0
RAILS_PGOOD	RAILS_PGOOD
SUS_S3_N	SUS_S3_N
SUS_S5_N	SUS_S5_N
RSMRST_OUT_N	RSMRST_OUT_N
PLTRST_N	PLTRST_N
THERMTRIP_N	THERMTRIP_N
CARRIER_HOT_ND	CARRIER_HOT_N
PROCHOT_L	PROCHOT_L
UART0	UART0
I2CO_SCL	I2CO_SCL
I2CO_SDA	I2CO_SDA
FAN_PWMOUT	FAN_PWMOUT
FAN_TACHIN	FAN_TACHIN
FCH_SPI	FCH_SPI
FCH_UART0	FCH_UART0
CLK_48M	CLK_48M
RTC_32K	RTC_32K
VBAT_RTC_IOB	VBAT_RTC_IOB
VBAT_RTC_CB	VBAT_RTC_CB

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC se

Sheet: /EC, firmware, clocks, RTC, debug/
File: ec_fw.kicad_sch

Title: MP62 AM5 CPU board: EC, firmware, clocks, RTC, debug

Size: A4	Date: 2026-10-04	Rev: A-am5-bd0
KiCad E.D.A. 9.0.2	Id: 8/11	

